Linear-Quadratic Systems

Name _____

Algebra 2 Lesson 2-7 (fast track) Date _____ Period _____

Learning goal: I can solve a linear-quadratic system by substitution and shade the solution region of a linear-quadratic system of inequalities.

Language goal: I can explain why my answer solves the system using the words substitute, intersection, boundary, and region.

On the quiz: (1) Count the solutions of a line and a parabola (full Savvas quiz only). (2) Substitute to solve a system whose two equations both start "$y =$" (bonus; Example 3's first step). (3) Find how far a ball travels before it lands on a hill: set the path equal to the hill and solve (Example 3). (4) Shade the solution region of one quadratic and one linear inequality (Example 4). (5) Solve with a calculator's intersection feature (full Savvas quiz only).

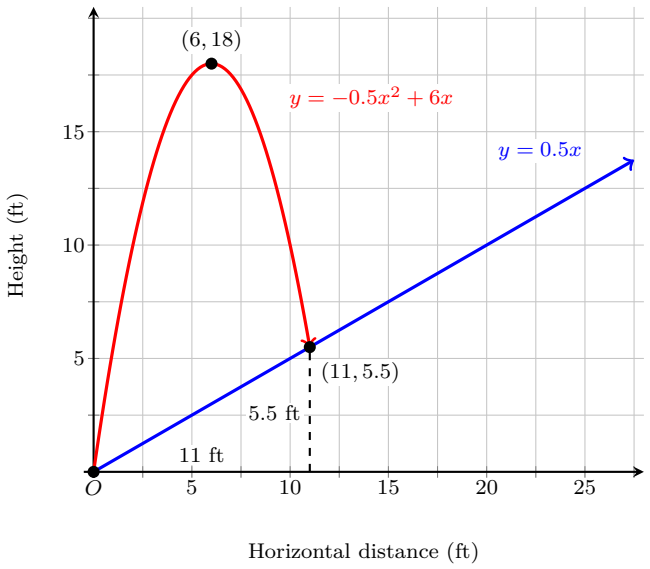
Example 3: Applying a Linear-Quadratic System

Andrew kicks a ball up a hill for his dog, Laika, to chase. The hill is modeled by a line through the origin, $y = 0.5x$. The path of the ball is modeled by the quadratic function $y = -0.5x^2 + 6x$. How far does the ball travel horizontally? How far must Laika run up the hill to catch it?

1. The ball lands where the path meets the hill: an **intersection**. Both equations give y, so **substitute** $0.5x$ for y:

$$\begin{aligned} 0.5x &= -0.5x^2 + 6x \\ 0 &= -0.5x^2 + 5.5x \\ 0 &= -0.5x(x - 11) \end{aligned}$$

$$x = 0 \quad \text{or} \quad x = 11$$



2. $x = 0$ is where Andrew kicks the ball. $x = 11$ is where it lands on the hill, at $y = 0.5(11) = 5.5$. **The ball travels 11 ft horizontally.**

3. Laika runs along the hill: the hypotenuse of a right triangle with legs 11 ft and 5.5 ft.

$$\sqrt{11^2 + 5.5^2} = \sqrt{151.25} \approx 12.3 \text{ ft}$$

Laika runs about 12.3 ft. (Answer *every* part of the question.)

Say it: I _____ $0.5x$ for y because at the _____ the path and the hill have the same y-value. The solution $x = 11$ means ...

Try It 3: Revenue and cost

Revenue for the high school band concert is given by the function $y = -30x^2 + 250x$, where x is the ticket price, in dollars. The cost of the concert is given by the function $y = 490 - 30x$. At what ticket price will the band make enough revenue to cover their costs?

a. Hint: set revenue equal to cost, then solve.

b. Hint: test a price between your two answers.

The band covers its costs for ticket prices from _____ to _____ dollars.

Practice: Bounce or splash? (DOK 3, Practice 29)

A golfer accidentally hits a ball toward a water hazard that is downhill from current position on the fairway. The hill can be modeled by a line through the origin with slope $-\frac{1}{8}$. The path of the ball can be modeled by the function $y = -\frac{1}{100}x^2 + \frac{3}{2}x$.

Part A If the golfer stands at the origin, and the water hazard is 180 yd away, will the golfer's ball bounce or splash?

Hill: $y =$ _____ Substitute and solve:

Part B How far did the ball land from the edge of the water hazard?

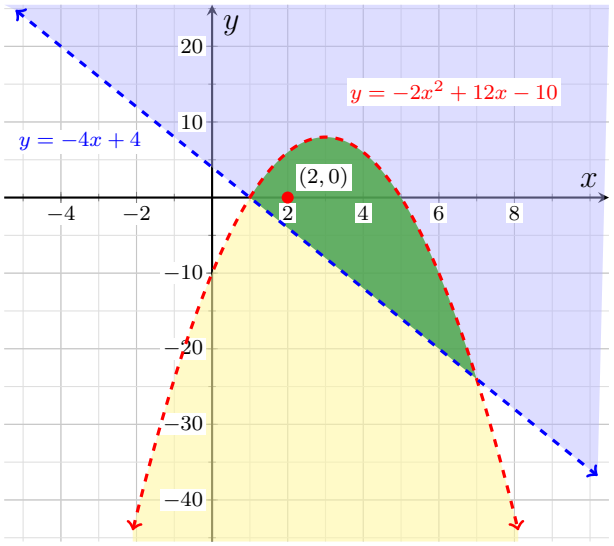
Part C Does it matter whether you measure the 180 yd horizontally or along the hill? Explain.

Example 4: Solve a Linear-Quadratic System of Inequalities

How can you solve this system of inequalities? $y < -2x^2 + 12x - 10$ and $4x + y > 4$

Step 1 Graph each **boundary**. "<" and ">": **dashed**, the boundary is not part of the solution.

- Parabola $y = -2x^2 + 12x - 10$: vertex $(3, 8)$, x-intercepts 1 and 5, y-intercept -10. Test $(0, 0)$: $0 < -10$? False. Shade the side *without* the origin: below the parabola.
- Line: $4x + y > 4$ is $y > -4x + 4$ (slope -4, y-intercept 4). Test $(0, 0)$: $0 > 4$? False. Shade above the line.



Step 2 The solution **region** is where the shadings overlap (darkest). The boundaries meet where $-2x^2 + 12x - 10 = -4x + 4$: $x^2 - 8x + 7 = 0$, $(x - 1)(x - 7) = 0$, at $(1, 0)$ and $(7, -24)$.

Check a point in the region, $(2, 0)$:

$0 < -2(2)^2 + 12(2) - 10$, $0 < 6$ ✓

$4(2) + 0 > 4$, $8 > 4$ ✓

Say it: The point $(2, 0)$ is in the solution _____ because ... Each _____ is dashed because ...

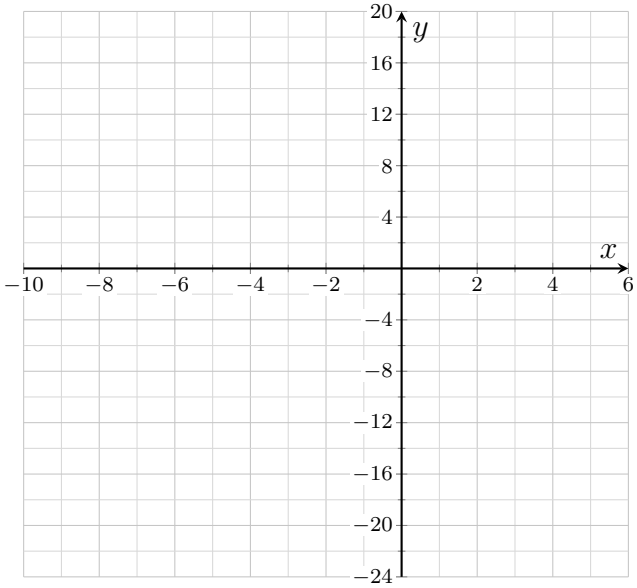
Try It 4: Shade the solution region

Solve the system of inequalities $y > x^2 + 6x - 12$, $3x - y \geq -8$ using shading.

1. Graph $y = x^2 + 6x - 12$, dashed or solid?
   Vertex: _____ y-intercept: _____
2. Solve $3x - y \geq -8$ for y (dividing by -1 flips the sign), dashed or solid?
   y _____
3. Test $(0, 0)$ in each inequality and shade.
4. Darken the region where the shadings overlap.
5. Check a point in your region in *both* original inequalities:

_____

_____



IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
HVZ ★	Solve a linear-quadratic system of equations: lines and parabolas	Quiz 2 (bonus) and 3: substitute, solve the quadratic, find both points (Example 3)
VQU ★	Solve a linear-quadratic system of equations by graphing: lines and parabolas	Quiz 4: graph the parabola and the line (the boundaries); Quiz 2 by graphing; Quiz 1 (full Savvas quiz only)
U5D ★	Solve systems of linear inequalities by graphing	Quiz 4: dashed or solid, test a point, shade the overlap (Example 4)
BW5	Solve a system of equations using substitution	Warm-up: the same move with two lines (Lesson 1-6)
M69	Solve a system of equations by graphing	Warm-up: read where two graphs meet (Lesson 1-5)

SmartScore 70 or above on each starred skill replaces your quiz score.

Exit question: Substitute and solve

Use substitution to solve the system of equations.

$$\begin{cases} y = -x^2 + x + 12 \\ y = 2x - 8 \end{cases}$$

Solutions: _____ and _____

Say it: The two solutions are the points of _____ of the line and the parabola. I checked one by ...

Before the quiz, finish these sentences.

To solve a linear-quadratic system, I _____ one expression for y into the other equation and solve the quadratic.

The solutions are the points of _____ of the line and the parabola: there can be 0, 1, or 2 of them.

A dashed _____ is not part of the solution; the solution _____ is where the shadings overlap.